Tomás Freire

tomas.fa.freire@gmail.com | Google Scholar | LinkedIn | GitHub

Profile

Applied mathematician combining mechanistic modeling, statistical inference and machine learning to study biological problems. My work ranges from modeling antibiotic resistance to analyzing clinical and microbiome data, including collaborations with experimental researchers.

Selected Research Projects

Predictive Modeling of Clinical State from Vaginal Microbiome Data

First-author paper: *Microbiome*, 2026 [Paper]

- Developed an interpretable model linking microbiome composition to bacterial vaginosis.
- Achieved 93.8% mean accuracy for binary BV classification, comparable to ML benchmarks.
- Identified context-dependent roles of microbial taxa, showing how their associations with vaginal health vary with community composition.

Spatial Modeling and Simulation of Antibiotic Resistance Dynamics

First-author paper: *PLOS Computational Biology*, 2024 [Paper]

- Developed and simulated PDE models of selection between antibiotic-sensitive and resistant bacteria in spatially heterogeneous environments.
- Derived a mathematical criterion predicting which resistant strains are favored by drug gradients, bacterial movement and habitat conditions.

Context-Dependent Selection in Multispecies Communities

First-author paper: *Bulletin of Mathematical Biology*, 2025 [Paper]

- Developed a mechanistic framework and used linear regression to estimate associations between gut microbiome composition and selection between resistant and susceptible *E. coli* strains in mice.

Experience

Doctoral Researcher *Sep. 2023 - Present*

Instituto Superior Técnico, Lisbon, Portugal

Teaching Assistant *Sep. 2023 - Jan. 2025*

Instituto Superior Técnico, Lisbon, Portugal

- Taught Calculus III and Probability & Statistics to undergraduate students.

Research Intern *Apr. 2022 - Aug. 2022*

University of Michigan - Biophysics Department, Ann Arbor, MI, USA

- Developed computational models of antibiotic resistance in an interdisciplinary lab, working alongside experimental researchers.

Technical Skills

Programming: Python (NumPy, pandas, SciPy, scikit-learn, XGBoost, Matplotlib, PyTorch, Streamlit); R, Mathematica; Git/GitHub.
Statistics & ML: Statistical inference, supervised and unsupervised learning, model interpretation and uncertainty quantification.
Modeling: ODE/PDE models, parameter estimation, numerical simulation, stability and bifurcation analysis.
Biological Data: Microbiome, metagenomic and clinical data analysis; data preprocessing and scientific visualization.

Education

PhD in Mathematics *2023 - Present*

Instituto Superior Técnico, University of Lisbon, Portugal

Thesis submitted October 2026; defense pending.

MSc & BSc in Applied Mathematics and Computation *2017 - 2022*

Instituto Superior Técnico, Lisbon

Communication & Awards

- Contributed talks at MEEHubs 2024 (Oxford, UK), Vaginal Microbiota and Genital Infection Dynamics 2025 (Paris, France), and ECMTB 2026 (Graz, Austria).
- Poster Award, DSABNS 2026.